STALKS COMMUTE WITH THE LEFT KAN EXTENSION ALONG A FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The description of the left Kan extension. Same chapter, the description of the left Kan extension as a colimit over morphisms into the images of objects. Supplies the first expression in the computation.
- The description of the stalk functor. Same chapter, Equation equation-stalk. Supplies the description of the stalks appearing in the computation.
- Colimits with connected fibres are unchanged. Categories, Lemma categories-lemma-colimit-constant-connected-fibers. Justifies the passage from the combined colimit to the colimit over the smaller index category.

Source reference. Original equation 04EH: sources/stacks-sites/pinned-original-source.tex, source lines 8735–8735. Navigation pointer invoked at proof line 8735; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05US: sources/stacks-sites/pinned-original-source.tex, source lines 8754–8754. Navigation pointer invoked at proof line 8754; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Let $v : \mathcal{D} \to Sets$ be a functor and set $w = v \circ u$. Denote q, resp., p the stalk functor (04EH) associated to v, resp. w. Then $(u_p\mathcal{F})_q = \mathcal{F}_p$ functorially in the presheaf $\mathcal{F}$ on $\mathcal{C}$.*

Proof. This is a simple categorical fact. We have

$$\begin{aligned}(u_p\mathcal{F})_q &= \operatorname{colim}_{(V,y)} \operatorname{colim}_{U,\phi:V\to u(U)} \mathcal{F}(U) \\ &= \operatorname{colim}_{(V,y,U,\phi:V\to u(U))} \mathcal{F}(U) \\ &= \operatorname{colim}_{(U,x)} \mathcal{F}(U) \\ &= \mathcal{F}_p\end{aligned}$$

The first equality holds by the definition of u_p and the definition of the stalk functor. Observe that $y \in v(V)$. In the second equality we simply combine colimits. To see the third equality we apply Categories, Lemma 05US to the functor F of diagram categories defined by the rule

$$F((V, y, U, \phi : V \to u(U))) = (U, v(\phi)(y)).$$

This makes sense because $w(U) = v(u(U))$. Let us check the hypotheses of Categories, Lemma 05US. Observe that F has a right inverse, namely $(U, x) \mapsto (u(U), x, U, \mathrm{id} : u(U) \to u(U))$. Again this makes sense because $x \in w(U) = v(u(U))$. On the other hand, there is always a morphism

$$(V, y, U, \phi : V \to u(U)) \longrightarrow (u(U), v(\phi)(y), U, \mathrm{id} : u(U) \to u(U))$$

in the fibre category over (U, x) which shows the fibre categories are connected. The fourth and final equality is clear. □